Long-range order and spontaneous symmetry breaking in quantum spin systems and hard core bosons on a lattice

part 3 LRO and SSB in the ground states of quantum spin systems with discrete symmetry

Advanced Topics in Statistical Physics
by Hal Tasaki

§ Ising model under transveres magnetic field

a simple theoretical example of quantum many-body system

Hamiltonian

$$\hat{H}_L = -\sum_{\{u,v\}\in\mathcal{B}_L} \hat{S}^{(z)}_u \hat{S}^{(z)}_v - \lambda \boxed{\sum_{u\in\Lambda_L} \hat{S}^{(x)}_u} \quad (1)$$

$= \hat{O}^{(x)}_L$

$\lambda \geq 0$ a fixed parameter

We focus on the ground state(s) of $\hat{H}_L$ for large L in any $d=1,2,\cdots$

symmetry

$\hat{T}_w$: translation $\qquad \hat{T}_w^\dagger \hat{H}_L \hat{T}_w = \hat{H}_L$ for any w (2)

$\hat{U} = \hat{U}^{(x,\pi)}_L$ (3) $\qquad \hat{U}^\dagger \hat{H}_L \hat{U} = \hat{H}_L$ (4)

$\hat{U}^2 = \hat{I}$ (5) $\qquad \mathbb{Z}_2$ symmetry

order operator

$\hat{O}^{(z)}_L = \sum_u \hat{S}^{(z)}_u$ (6) $\qquad \hat{U}^\dagger \hat{O}^{(z)}_L \hat{U} = -\hat{O}^{(z)}_L$ (7)

▷ trivial case with $\lambda = 0$

easy because $[\hat{H}_L, \hat{O}_L^{(2)}] = 0$ (1)

doubly degenerate exact ground states

$$|\Phi_L^{\uparrow}\rangle = \bigotimes_{u \in \Lambda_L} |\uparrow\rangle_u \quad (2) \qquad |\Phi_L^{\downarrow}\rangle = \bigotimes_{u \in \Lambda_L} |\downarrow\rangle_u \quad (3)$$

$$\hat{H}_L |\Phi_L^{\uparrow}\rangle = E_L^{GS} |\Phi_L^{\uparrow}\rangle \quad (4) \qquad \hat{H}_L |\Phi_L^{\downarrow}\rangle = E_L^{GS} |\Phi_L^{\downarrow}\rangle \quad (5)$$

with $E_L^{GS} = -\frac{1}{4} |\mathcal{B}_L|$ (6)

$\mathbb{Z}_2$ symmetry is explicitly broken

$$\hat{O}_L^{(2)} |\Phi_L^{\uparrow}\rangle = \frac{L^d}{2} |\Phi_L^{\uparrow}\rangle \quad (7) \qquad \hat{O}_L^{(2)} |\Phi_L^{\downarrow}\rangle = -\frac{L^d}{2} |\Phi_L^{\downarrow}\rangle \quad (8)$$

exact ground states $|\Phi_L^{\uparrow}\rangle, |\Phi_L^{\downarrow}\rangle$ are physical ground states with LRO and symmetry breaking

▷ case with $\lambda > 0$

nontrivial and interesting because $[\hat{H}_L, \hat{O}_L^{(2)}] \neq 0$ (1)

theorem for any L, the ground state of $\hat{H}_L$ is unique and written as

$$|\Phi_L^{GS}\rangle = \sum_{\sigma \in \mathcal{S}}' \alpha_\sigma |\sigma\rangle \quad (2) \quad \text{with } \alpha_\sigma > 0 \text{ for any } \sigma \in \mathcal{S}$$

proof:
- $\langle\sigma|\hat{H}_L|\tau\rangle \in \mathbb{R}$, $\langle\sigma|\hat{H}_L|\tau\rangle \leq 0$ if $\sigma \neq \tau$
- any $\sigma, \tau \in \mathcal{S}$ are "connected" via nonzero $\langle\sigma'|\hat{H}_L|\sigma''\rangle$

Perron-Frobenius theorem (for real symmetric matrices) implies the theorem

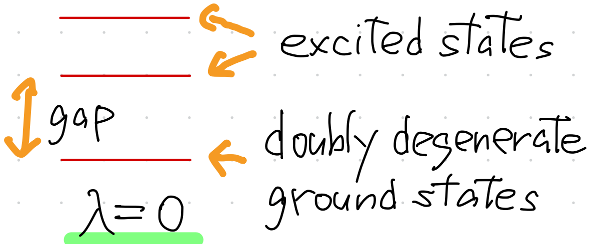
uniqueness + symmetry of $\hat{H}_L \rightarrow \hat{U}|\Phi_L^{GS}\rangle = c_L|\Phi_L^{GS}\rangle$ with $|c_L| = 1$ (2)

$$\langle\Phi_L^{GS}|\hat{O}_L^{(2)}|\Phi_L^{GS}\rangle = \langle\Phi_L^{GS}|\hat{U}^\dagger\hat{O}_L^{(2)}\hat{U}|\Phi_L^{GS}\rangle = -\langle\Phi_L^{GS}|\hat{O}_L^{(2)}|\Phi_L^{GS}\rangle \quad (3)$$

$$\langle\Phi_L^{GS}|\hat{O}_L^{(2)}|\Phi_L^{GS}\rangle = 0 \quad (4)$$

NO symmetry breaking

iii) case with $0<\lambda\ll 1$ perturbation to the trivial model with $\lambda=0$

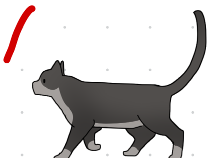


$E_L^{1st}-E_L^{GS}\sim e^{-cL^d}$ exponentially small splitting

$$|\Phi_L^{GS}\rangle \simeq \frac{1}{\sqrt{2}}\{|\Phi_L^{\uparrow}\rangle+|\Phi_L^{\downarrow}\rangle\} \quad (1)$$ unique exact ground state

$$|\Phi_L^{1st}\rangle \simeq \frac{1}{\sqrt{2}}\{|\Phi_L^{\uparrow}\rangle-|\Phi_L^{\downarrow}\rangle\} \quad (2)$$ low-lying excited state

apparently Schrödinger's cats **unphysical**



they must collapse into $|\Phi_L^{\uparrow}\rangle$ or $|\Phi_L^{\downarrow}\rangle$

since $$(\hat{O}_L^{(z)})^2|\Phi_L^{\uparrow}\rangle=\left(\frac{L^d}{2}\right)^2|\Phi_L^{\uparrow}\rangle \quad (3) \qquad (\hat{O}_L^{(z)})^2|\Phi_L^{\downarrow}\rangle=\left(\frac{L^d}{2}\right)^2|\Phi_L^{\downarrow}\rangle \quad (4)$$

$$\langle\Phi_L^{GS}|\left(\frac{\hat{O}_L^{(z)}}{L^d}\right)^2|\Phi_L^{GS}\rangle \simeq \frac{1}{2}\left\{\langle\Phi_L^{\uparrow}|\left(\frac{\hat{O}_L^{(z)}}{L^d}\right)^2|\Phi_L^{\uparrow}\rangle+\langle\Phi_L^{\downarrow}|\left(\frac{\hat{O}_L^{(z)}}{L^d}\right)^2|\Phi_L^{\downarrow}\rangle\right\}=\frac{1}{4} \quad (5)$$

long-range order!

illustration by Chisato Natuse

recalling $\langle \Phi_L^{GS} | \hat{O}_L^{(2)} | \Phi_L^{GS} \rangle = 0$ (1) (no symmetry breaking)

we see

$$\boxed{\sqrt{\langle \Phi_L^{GS} | \left(\frac{\hat{O}_L^{(2)}}{L^d} \right)^2 | \Phi_L^{GS} \rangle - \left(\langle \Phi_L^{GS} | \frac{\hat{O}_L^{(2)}}{L^d} | \Phi_L^{GS} \rangle \right)^2}} = \frac{1}{2} \quad (2)$$

$\parallel$ fluctuation of $\hat{O}_L^{(2)} / L^d$ in $|\Phi_L^{GS}\rangle$

The density of a macroscopic quantity has nonzero fluctuation

the exact ground state $|\Phi_L^{GS}\rangle$ with LRO but without SSB is unphysical as a state of a macroscopic system

what are physical ground states?

two ground states because $\mathbb{Z}_2$ symmetry is broken

$|\Phi_L^{\uparrow}\rangle$ and $|\Phi_L^{\downarrow}\rangle$ are macroscopic ground states with both LRO and SSB

▨ macroscopic ground state (general definition)

$\hat{H}_L$ translation-invariant Hamiltonian on Λ_L with short-range interactions

ground state energy density $\quad \epsilon^{GS} = \lim_{L\uparrow\infty} \dfrac{E^L_{GS}}{L^d} \quad$ (1)

$$\hat{T}_w |\Psi_L\rangle = \text{const.} |\Psi_L\rangle \qquad (2)$$

if $|\Psi_L\rangle$ is translation invariant and satisfies

$$\lim_{L\uparrow\infty} \frac{1}{L^d} \langle \Psi_L | \hat{H}_L | \Psi_L \rangle = \epsilon^{GS} \qquad (3)$$

we say $|\Psi_L\rangle$ (for large L) is a macroscopic ground state

infinite volume state $\omega(\cdot)$ defined by $\omega(\hat{A}) = \lim_{L\uparrow\infty} \langle \Psi_L | \hat{A} | \Psi_L \rangle$ is a ground state in the sense of the operator algebraic formulation of quantum spin systems. → Appendix A of Koma-Tasaki 1994

▷ back to the model with $0<\lambda\ll 1$

unphysical

$|\Phi_L^{GS}\rangle, |\Phi_L^{1st}\rangle$ macroscopic ground states with LRO but without SSB

$|\Phi_L^{\uparrow}\rangle, |\Phi_L^{\downarrow}\rangle$ macroscopic ground states with LRO and SSB

physical

▷ hints for general theories

$$\hat{O}_L^{(z)}|\Phi_L^{GS}\rangle \simeq \hat{O}_L^{(z)}\frac{1}{\sqrt{2}}\{|\Phi_L^{\uparrow}\rangle+|\Phi_L^{\downarrow}\rangle\}$$
$$\simeq \frac{L^d}{2}\frac{1}{\sqrt{2}}\{|\Phi_L^{\uparrow}\rangle-|\Phi_L^{\downarrow}\rangle\} \simeq \frac{L^d}{2}|\Phi_L^{1st}\rangle \qquad (1)$$

this suggests

$$|\Phi_L^{\uparrow}\rangle \simeq \frac{1}{\sqrt{2}}\left\{|\Phi_L^{GS}\rangle+|\Phi_L^{1st}\rangle\right\} \simeq \frac{1}{\sqrt{2}}\left\{|\Phi_L^{GS}\rangle+\frac{\hat{O}_L^{(z)}|\Phi_L^{GS}\rangle}{\|\hat{O}_L^{(z)}|\Phi_L^{GS}\rangle\|}\right\} \qquad (2)$$

$$|\Phi_L^{\downarrow}\rangle \simeq \frac{1}{\sqrt{2}}\left\{|\Phi_L^{GS}\rangle-|\Phi_L^{1st}\rangle\right\} \simeq \frac{1}{\sqrt{2}}\left\{|\Phi_L^{GS}\rangle-\frac{\hat{O}_L^{(z)}|\Phi_L^{GS}\rangle}{\|\hat{O}_L^{(z)}|\Phi_L^{GS}\rangle\|}\right\} \qquad (3)$$

§ general theories for low-lying states and SSB

Horsch, von der Linden 1988, Kaplan, Horsch, von der Linden 1989

▷ setting

$\hat{H}_L$ translation-invariant Hamiltonian on Λ_L with short-range interactions

$\hat{O}_L = \sum_{u\in\Lambda_L} \hat{o}_u$ (1) order operator ($\hat{o}_u = \hat{o}_u^\dagger$ acts only on u, $\hat{o}_u = \hat{T}_u \hat{o}_0 \hat{T}_u^\dagger$)

$|\Phi_L^{GS}\rangle$ exact translation-invariant ground state of $\hat{H}_L$

E_L^{GS} ground state energy and its density $\epsilon^{GS} = \lim_{L\uparrow\infty} \frac{E_L^{GS}}{L^d}$ (2)

▷ assumptions

$\left(\langle\Phi_L^{GS}|\hat{o}_u \hat{o}_v|\Phi_L^{GS}\rangle \simeq q_0 \text{ for large } |u-v| \right)$

$$\langle\Phi_L^{GS}|\left(\frac{\hat{O}_L}{L^d}\right)^2|\Phi_L^{GS}\rangle \geq q_0 > 0 \quad (3)$$

there is LRO

$$\langle\Phi_L^{GS}|\left(\frac{\hat{O}_L}{L^d}\right)^m|\Phi_L^{GS}\rangle = 0 \text{ for } m=1,3 \quad (4)$$

but no SSB

▷ low-lying states

see P.7-(1)

define $$|\Gamma_L\rangle = \frac{\hat{O}_L|\Phi_L^{GS}\rangle}{\|\hat{O}_L|\Phi_L^{GS}\rangle\|} \quad (1)$$

theorem (Horsch, von der Linden 1988)

$|\Gamma_L\rangle$ is orthogonal to $|\Phi_L^{GS}\rangle$ there is a constant $C>0$ s.t.

$$0 \leqslant \langle\Gamma_L|\hat{H}_L|\Gamma_L\rangle - E_L^{GS} \leqslant \frac{C}{L^d} \quad (2)$$

corollary there is an energy eigenstate $|\Psi_L\rangle$ s.t. $\langle\Phi_L^{GS}|\Psi_L\rangle = 0$

with eigenvalue E_L^1 satisfying $0 \leqslant E_L^1 - E_L^{GS} \leqslant \frac{C}{L^d}$ (3) (proof: easy)

then $$\lim_{L\uparrow\infty} \frac{1}{L^d}\langle\Gamma_L|\hat{H}_L|\Gamma_L\rangle = \lim_{L\uparrow\infty} \frac{1}{L^d}\langle\Psi_L|\hat{H}|\Psi_L\rangle = \epsilon^{GS} \quad (4)$$

$|\Gamma_L\rangle$ and $|\Psi_L\rangle$ are macroscopic ground states

chose $|\Psi_L\rangle$ to be translation invariant proof of theorem

$$|\Gamma_L\rangle = \frac{\hat{O}_L|\Phi_L^{GS}\rangle}{\|\hat{O}_L|\Phi_L^{GS}\rangle\|} \qquad (1)$$

p8-(4)

$$\langle\Phi_L^{GS}|\Gamma_L\rangle = \frac{\langle\Phi_L^{GS}|\hat{O}_L|\Phi_L^{GS}\rangle}{\|\hat{O}_L|\Phi_L^{GS}\rangle\|} = 0 \qquad (2)$$

we write $\langle\cdots\rangle = \langle\Phi_L^{GS}|\cdots|\Phi_L^{GS}\rangle$ (3)

$$\langle\Gamma_L|\hat{H}_L|\Gamma_L\rangle = \frac{\langle\Phi_L^{GS}|\hat{O}_L\hat{H}_L\hat{O}_L|\Phi_L^{GS}\rangle}{\langle\Phi_L^{GS}|\hat{O}_L\hat{O}_L|\Phi_L^{GS}\rangle} = \frac{\langle OHO\rangle}{\langle O^2\rangle} \qquad (4)$$

$\hat{O}_L^\dagger = O_L$

$$\langle\Gamma_L|\hat{H}_L|\Gamma_L\rangle - E_L^{GS} = \frac{\langle OHO\rangle - E_L^{GS}\langle O^2\rangle}{\langle O^2\rangle}$$

we shall evaluate this double commutator!

$$= \frac{\langle OHO\rangle - \frac{1}{2}\langle O^2H\rangle - \frac{1}{2}\langle HO^2\rangle}{\langle O^2\rangle} = \frac{\langle[O,[H,O]]\rangle}{2\langle O^2\rangle} \qquad (5)$$

$$\hat{H}_L = \sum_{u\in\Lambda_L} \hat{h}_u \quad (1)$$ $\hat{h}_u$ acts only on sites v s.t. $|u-v|\le r_0$

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$$\hat{O}_L = \sum_{u\in\Lambda_L} \hat{O}_u \quad (2)$$ $\hat{O}_u$ acts only on site u

$$[\hat{O}_L,[\hat{H}_L,\hat{O}_L]] = \sum_{u,v,w\in\Lambda_L} [\hat{O}_u,[\hat{h}_v,\hat{O}_w]] = \sum_{v\in\Lambda_L} \sum_{\substack{u,w\\(|u-v|\le r_0\\ |w-v|\le r_0)}} [\hat{O}_u,[\hat{h}_v,\hat{O}_w]] \quad (3)$$

$$\|[\hat{O}_L,[\hat{H}_L,\hat{O}_L]]\| \le \sum_{v\in\Lambda_L} \sum_{\substack{u,w\\(|u-v|\le r_0\\ |w-v|\le r_0)}} \underbrace{\|[\hat{O}_u,[\hat{h}_v,\hat{O}_w]]\|}_{\le 4\|\hat{h}_0\|\,\|\hat{O}_0\|^2} \le A L^d \quad (4)$$

operator norm (see next page)

$$\langle[\hat{O},[\hat{H},\hat{O}]]\rangle \le A L^d \quad (5)$$

p8-(3) → $\langle \hat{O}^2\rangle \ge q_0 L^{2d}$ (6) ← LRO!!

$$\langle\Gamma_L|\hat{H}_L|\Gamma_L\rangle - E_L^{GS} = \frac{\langle[[O,[H,O]]\rangle}{2\langle O^2\rangle} \le \frac{A L^d}{2 q_0 L^{2d}} = \frac{C}{L^d} \quad (7)$$

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operator norm

$\mathcal{H}$ finite dimensional Hilbert space

$\hat{A}$ operator on $\mathcal{H}$

$$\|\hat{A}\| = \max_{\substack{|\Phi\rangle \in \mathcal{H} \\ (\|\Phi\|=1)}} \|\hat{A}|\Phi\rangle\| \qquad (1)$$

$$\|\hat{A}+\hat{B}\| \leq \|\hat{A}\| + \|\hat{B}\| \qquad (2)$$

$$\|\hat{A}\hat{B}\| \leq \|\hat{A}\|\,\|\hat{B}\| \qquad (3)$$

$$|\langle\Phi|\hat{A}|\Phi\rangle| \leq \|\hat{A}\|\,\langle\Phi|\Phi\rangle \qquad (4)$$

if $\hat{A} = \hat{A}^\dagger$ with eigenvalues $a_1, \cdots, a_D$

$$\|A\| = \max_j |a_j| \qquad (5)$$

III macroscopic ground states with LRO and SSB

p7-(2),(3)

define $|\Xi_L^{\pm}\rangle = \frac{1}{\sqrt{2}}\{|\Phi_L^{GS}\rangle \pm |\Gamma_L\rangle\}$ (1)

clearly $\||\Xi_L^{\pm}\|| = 1$ (2) $\quad \langle\Xi_L^{+}|\Xi_L^{-}\rangle = 0$ (3)

note also $\langle\Xi_L^{\pm}|\hat{H}_L|\Xi_L^{\pm}\rangle = \frac{1}{2}\{\langle\Phi_L^{GS}|H_L|\Phi_L^{GS}\rangle + \langle\Gamma_L|\hat{H}_L|\Gamma_L\rangle\}$

$$\leq \frac{1}{2}\{E_L^{GS} + E_L^{GS} + C/L^d\} = E_L^{GS} + C/(2L^d)$$

$|\Xi_L^{\pm}\rangle$ are macroscopic ground states

$$\langle\Xi_L^{+}|\hat{O}_L|\Xi^{+}\rangle = \frac{1}{2}\left(\langle\Phi_L^{GS}| + \frac{\langle\Phi_L^{GS}|\hat{O}_L}{\|\hat{O}_L|\Phi_L^{GS}\rangle\|}\right)\hat{O}_L\left(|\Phi_L^{GS}\rangle + \frac{\hat{O}_L|\Phi_L^{GS}\rangle}{\|\hat{O}_L|\Phi_L^{GS}\rangle\|}\right)$$

p8-(4) p8-(3)

$$= \frac{\langle\Phi_L^{GS}|(\hat{O}_L)^2|\Phi_L^{GS}\rangle}{\|\hat{O}_L|\Phi_L^{GS}\rangle\|} = \sqrt{\langle\Phi_L^{GS}|(\hat{O}_L)^2|\Phi_L^{GS}\rangle} \geq \sqrt{q_0}\, L^d$$

$$\boxed{\pm\langle\Xi_L^{\pm}|\frac{\hat{O}_L}{L^d}|\Xi_L^{\pm}\rangle \geq \sqrt{q_0}}$$

$|\Xi_L^{\pm}\rangle$ exhibits (spontaneous) symmetry breaking!

($|\Xi_L^{\pm}\rangle$ may or may not a physical state with small fluctuation
See Appendix B, Koma Tasaki 1994)

Griffiths-type theorem

$$\hat{H}_L^h = \hat{H}_L - h\hat{O}_L \quad (1)$$

$|\Phi_L^{GS,h}\rangle$ an exact ground state of $\hat{H}_L^h$

h symmetry breaking field

Theorem (Kaplan, Horsch, von der Linden 1989)

$$\lim_{h\downarrow 0}\lim_{L\uparrow\infty}\langle\Phi_L^{GS,h}|\frac{\hat{O}_L}{L^d}|\Phi_L^{GS,h}\rangle \geq \sqrt{q_0} > 0 \quad (2)$$

part 2-p.4

infinitessimal symmetry breaking field leads to a ground state with SSB

proof

$$\langle\Xi_L^+|\hat{H}_L^h|\Xi_L^+\rangle \geq \langle\Phi_L^{GS,h}|\hat{H}_L^h|\Phi_L^{GS,h}\rangle \quad (3)$$

defintion of a ground state!

divide by hL^d

$$\langle\Phi_L^{GS,h}|\frac{\hat{O}_L}{L^d}|\Phi_L^{GS,h}\rangle \geq \langle\Xi_L^+|\frac{\hat{O}_L}{L^d}|\Xi_L^+\rangle + \frac{1}{hL^d}\left\{\langle\Phi_L^{GS,h}|\hat{H}_L|\Phi_L^{GS,h}\rangle - \langle\Xi_L^+|\hat{H}_L|\Xi_L^+\rangle\right\}$$

$$\geq \sqrt{q_0} + \underbrace{\frac{1}{hL^d}\left\{E_L^{GS} - \left(E_L^{GS} + \frac{C}{2L^d}\right)\right\}}_{\to 0 \text{ as } L\uparrow\infty} \quad (4)$$

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